

```

he_boss@LAPTOP-FGDPB110:~$ ls
SSP Week3 process process.c process.c.save process.c.save.1
he_boss@LAPTOP-FGDPB110:~$ pwd
/home/the_boss
he_boss@LAPTOP-FGDPB110:~$ data
command 'data' not found, did you mean:
command 'date' from deb coreutils (9.4-3ubuntu2)
command 'dat' from deb liballegro4-dev (2:4.4.3.1-4)
ry: sudo apt install <deb name>
he_boss@LAPTOP-FGDPB110:~$ whoami
he_boss
he_boss@LAPTOP-FGDPB110:~$ uname
linux
he_boss@LAPTOP-FGDPB110:~$ uname -r
4.18.33.2-microsoft-standard-WSL2
he_boss@LAPTOP-FGDPB110:~$ id
uid=1000(the_boss) gid=1000(the_boss) groups=1000(the_boss),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),38(dip),44(video),46(plugdev),100(users),107(netdev)
he_boss@LAPTOP-FGDPB110:~$ tty
dev/pts/0
he_boss@LAPTOP-FGDPB110:~$ uptime
05:55:35 up 2 min, 1 user, load average: 0.02, 0.02, 0.00
he_boss@LAPTOP-FGDPB110:~$ users
he_boss
he_boss@LAPTOP-FGDPB110:~$ who
he_boss pts/1 2026-08-08 05:53
he_boss@LAPTOP-FGDPB110:~$ w
05:55:35 up 2 min, 1 user, load average: 0.01, 0.02, 0.00
SSR TTY FROM LOGIN# IDLE JCPU PCPU WHAT
he_boss pts/1 05:53 2:0s 0.06s 0.06s -bash
he_boss@LAPTOP-FGDPB110:~$

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comments

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the_boss@DESKTOP-I8CBAJG:~$ groups
the_boss adm dialout cdrom floppy sudo audio dip video plugdev users netdev
the_boss@DESKTOP-I8CBAJG:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=the_boss
PWD=/home/the_boss
LOGNAME=the_boss
HOME=/home/the_boss
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/334_interop
LC_ALL=C.UTF-8:di=01;36:ln=01;36:mh=00;pi=40;33:o=01;35:do=01;35:bd=40;33:01=cd=40;33:0
LS_COLORS=33;11:n=00;su=37;41:sg=30;43:ca=00;tw=30;42:ow=30;42:st=37;44:ex=01;32:*target;
*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taZ=01;31:*\.*tma=01;31:*.l1z4=01;31:\
*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.z=01;31:*.Z=01;31:*.dz=01;31:*.gz=01;
31:*.lz=01;31:*.lzo=01;31:*.tz=01;31:*.bz2=01;31:*.bz=01;31:*.tbz2=01;31:*.tbz=01;31:*.tb
z=01;31:*.zip=01;31:*.zoo=01;31:*.zst=01;31:*.7z=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31
:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rm=01;31:*.rmi=01;31:*.rpm=01;31:*.deb=01;31:*.dsc=0
;31:*.pkg=01;31:*.efi=01;37:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.png=01;35:*.gif=01;35:
*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tif=01;35:*.tiff=01;35:*.xpm=01;35:*.xpm=
01;35:*.tga=01;35:*.icb=01;35:*.vda=01;35:*.vst=01;35:*.webp=01;35:*.ave=01;35:*.avi=01;35:
*.divx=01;35:*.flc=01;35:*.gif=01;35:*.h261=01;35:*.h263=01;35:*.h264=01;35:*.jpg=01;35:*.j
peg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.mpg=01;35:*.mpeg=01;35:*.mpe=01;35:*.mp4=01;35:*.m4v
=01;35:*.mov=01;35:*.mkv=01;35:*.ogm=01;35:*.ogv=01;35:*.qt=01;35:*.ts=01;35:*.vob=01;35:*.
wmv=01;35:*.xsm=01;35:*.xpm=01;35:*.xwd=01;35:
LESSCLOSE=/usr/bin/lesspipe %s %s
LESSOPEN=| /usr/bin/lesspipe %s
USER=the_boss
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/napd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/w
sl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracl
e/Java/javapath:/mnt/c/Windows/System32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32
/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/
Git/cmd:/mnt/c/Users/the_boss/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/the_boss/AppData/Local/Micros
oft/WindowsApps:/mnt/c/Users/the_boss/.dotnet/tools:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
the_boss@DESKTOP-I8CBAJG:~$

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Mem Swap: 2048.0 total, 2048.0 free, 0.0 used, 7221.0 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
386	teja	20	0	20324	11576	9456	S	2.4	0.1	0:00.35	systemd
1	root	20	0	21876	13100	9736	S	0.0	0.2	0:01.71	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
61	root	19	-1	50448	16196	15128	S	0.0	0.2	0:00.36	systemd-journal
108	root	20	0	25284	6672	5172	S	0.0	0.1	0:00.43	systemd-udev
126	systemd+	20	0	21344	13216	11092	S	0.0	0.2	0:00.26	systemd-resolve
127	systemd+	20	0	91036	7984	7016	S	0.0	0.1	0:00.18	systemd-timesyn
188	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.01	cron
191	message+	20	0	9644	5244	4540	S	0.0	0.1	0:00.14	dbus-daemon
198	root	20	0	17980	8884	7848	S	0.0	0.1	0:00.23	systemd-logind
201	root	20	0	1756628	13776	11444	S	0.0	0.2	0:00.23	wsl-pro-service
217	syslog	20	0	222516	5924	4640	S	0.0	0.1	0:00.15	rsyslogd
221	root	20	0	3124	1968	1836	S	0.0	0.0	0:00.01	agetty
231	root	20	0	107036	23216	13760	S	0.0	0.3	0:00.33	unattended-upgr
333	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
334	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.04	Relay(335)
335	teja	20	0	6080	5404	3688	S	0.0	0.1	0:00.14	bash
336	root	20	0	6696	4652	3876	S	0.0	0.1	0:00.02	login
387	teja	20	0	21168	3596	1868	S	0.0	0.0	0:00.00	(sd-pam)
416	teja	20	0	6080	5348	3676	S	0.0	0.1	0:00.06	bash
550	teja	20	0	9300	5684	3496	R	0.0	0.1	0:00.03	top

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the_boss@DESKTOP-I8C8AJG:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
none            3970372         0   3970372   0% /usr/lib/modules/6.18.33
.2-microsoft-standard-WSL2      4   3970368   1% /mnt/wsl
none            3970372        36   3970336   1% /mnt/wslg
drivers         498303300 134295152 364008148 27% /usr/lib/wsl/drivers
/dev/sdd        1055762868 1995676 1000063720 1% /mnt/c
none            3970372         0   3970372   0% /usr/lib/wsl/lib
rootfs          3964616      2772   3961844   1% /init
none            3970372        516   3969856   1% /run
none            3970372         0   3970372   0% /run/lock
none            3970372         0   3970372   0% /run/shm
none            3970372        80   3970292   1% /mnt/wslg/versions.txt
none            3970372        80   3970292   1% /mnt/wslg/doc
C:\            498303300 134295152 364008148 27% /mnt/c
tmpfs           794072        20    794052   1% /run/user/1000
the_boss@DESKTOP-I8C8AJG:~$ free
              total        used        free      shared  buff/cache   avail
Mem:           7940748      553896      6686120        3568       855128       738
Swap:          2097152           0      2097152

the_boss@DESKTOP-I8C8AJG:~$ ps
  PID TTY          TIME CMD
  335 pts/0      00:00:00 bash
  549 pts/0      00:00:00 ps

the_boss@DESKTOP-I8C8AJG:~$ top
top - 05:58:32 up 5 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,   0.1 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :   7754.6 total,   652.6 free,   546.4 used,   835.1 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7203.2 avail Mem
the_boss@DESKTOP-I8C8AJG:~$

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```

48 cal
49 hostname
50 uname -a
51 exit
52 sudo apt update
53 sudo apt upgrade
54 sudo apt install gcc
55 gcc --version
56 nano process.c
57 gcc process.c -o process
58 ./process
59 ls
60 pwd
61 date
62 cal
63 whoami
64 hostname
65 uname
66 uname -a
67 ps
68 top
69 history
70 Ctrl + O
71 mkdir Week3
72 cd Week3
73 nano task1.c
74 Ctrl + O~
75 hello
76 mkdir OSSP
77 cd OSSP
78 mkdir src include obj bin docs tests scripts assets
79 mkdir OSSP
80 cd OSSP
81 mkdir src include obj bin docs tests scripts assets
82 ls
83 ls -l
84 find . -maxdepth 3
85 touch .gitignore README.md LICENSE Makefile
86 touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c

```

```

the_boss@DESKTOP-1BC8A3G:~$ sudo apt install sysstat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
sysstat is already the newest version (12.7.6-1ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
the_boss@DESKTOP-1BC8A3G:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1335U
CPU family:             6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               3
BogoMIPS:               4992.02
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse
                        sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
                        nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic
                        movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch
                        ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2
                        smep bmi2 erms invpcid rdseed adx snap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves
                        avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize
                        ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:        VT-x
Hypervisor vendor:     Microsoft
Virtualization type:   full
Caches (sum of all):
  L1d:                  288 KiB (6 instances)
  L1i:                  192 KiB (6 instances)
  L2:                   7.5 MiB (6 instances)
  L3:                   12 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):    0-11
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
MDS:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Retbleed:              Not affected
Spec store bypass:     Not affected
Spectre v1:            Not affected
Spectre v2:            Not affected
Srbds:                 Not affected

```

```
94 git status
95 git add .
96 git status
97 git commit -m "Initial OSSP project structure"
98 git status
99 nothing to commit, working tree clean
100 nano fork_example.c
101 gcc fork_example.c -o fork_example
102 ./fork_example
103 ls
104 pwd
105 data
106 whoami
107 uname
108 uname -r
109 id
110 tty
111 uptime
112 users
113 who
114 w
115 groups
116 env
117 printenv
118 df
119 free
120 ps
121 top
122 df
123 free
```

```
103  ls
104  pwd
105  data
106  whoami
107  uname
108  uname -r
109  id
110  tty
111  uptime
112  users
113  who
114  w
115  groups
116  env
117  printenv
118  df
119  free
120  ps
121  top
122  df
123  free
124  ps
125  top
126  free
127  ps
128  top
129  vmstatvmsat
130  vmstat
131  iostat
132  lscpu
133  history
```

```
127  ps
128  top
129  vmstatvmsat
130  vmstat
131  iostat
132  lscpu
133  history
```

```
the_boss@DESKTOP-I8C8AJG: ~$ echo hello
hello
the_boss@DESKTOP-I8C8AJG:~$ bc
bc 1.07.1
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Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
^factor
0
factor 12
(standard_in) 2: syntax error
yes
3
8\mp
8
the_boss@DESKTOP-I8C8AJG:~$
```